PROYECTO PRÁCTICAS AULA 24

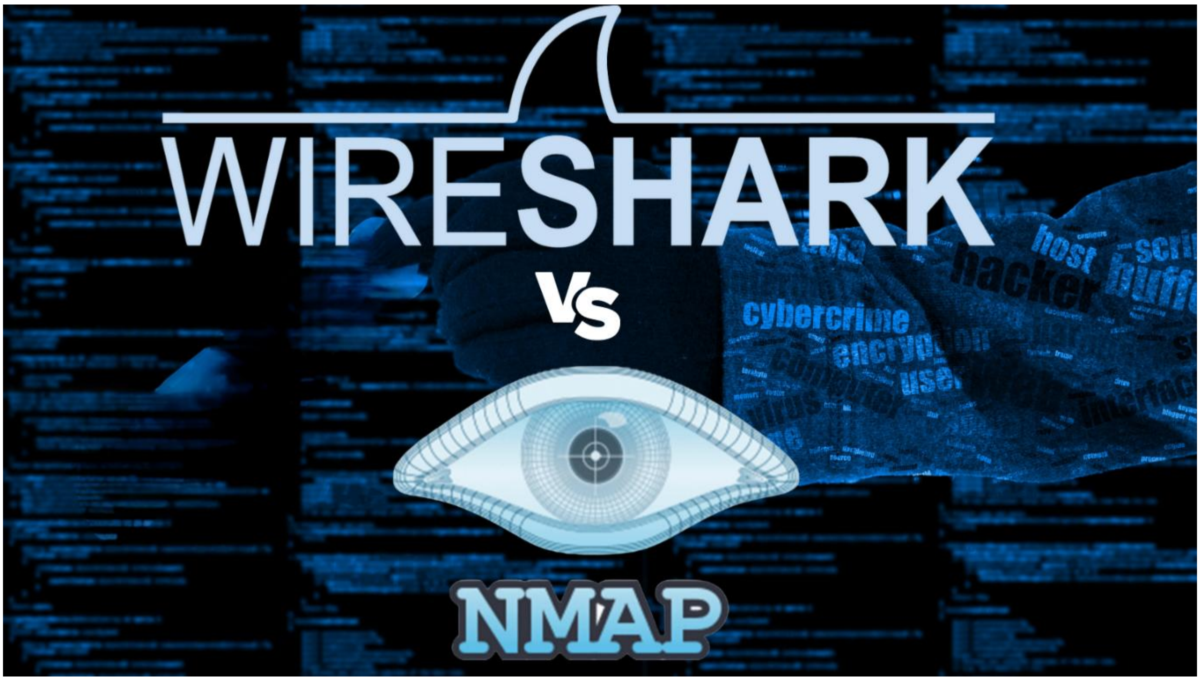
Grupo 5: Ivan, Santiago, Sergio R.

Grupo 6: Yuneida, Cristina.

Grupo 7: Sergio V, Felipe y Fernando.

Ciclo Formativo de Grado Superior en Administración de Sistemas Informáticos en Red (ASIR) – Año 2025/2026.

WIRESHARK Y NMAP

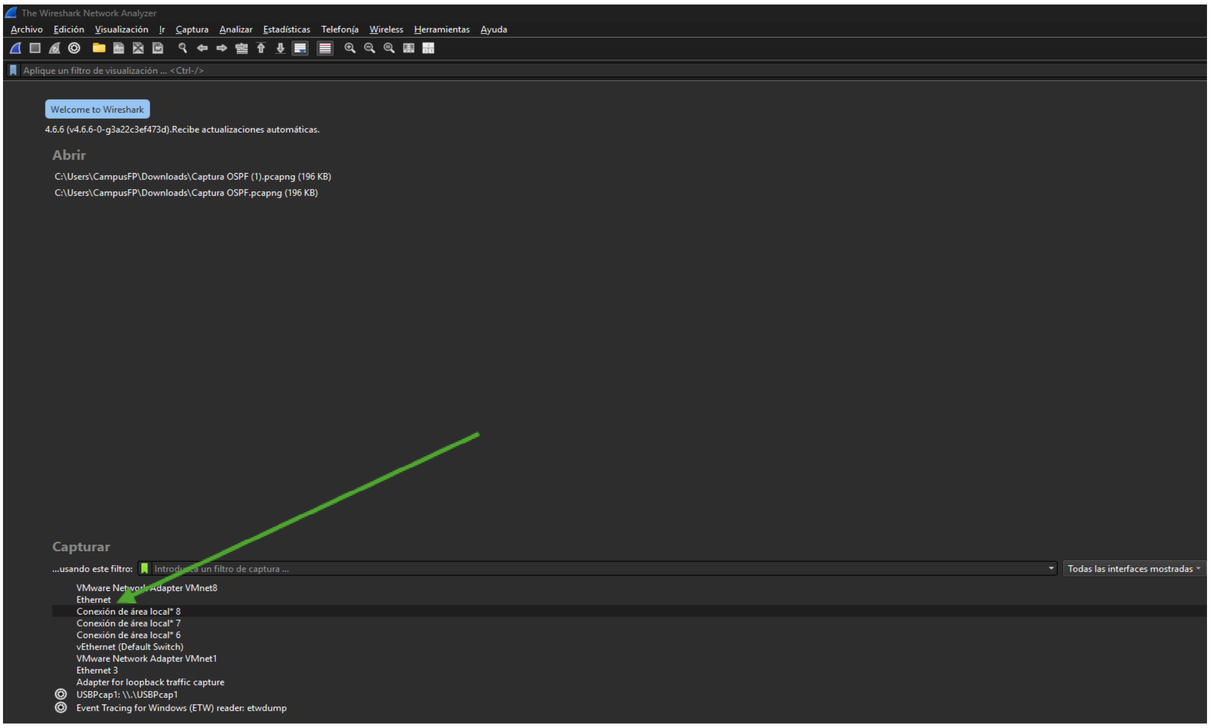


Índice:

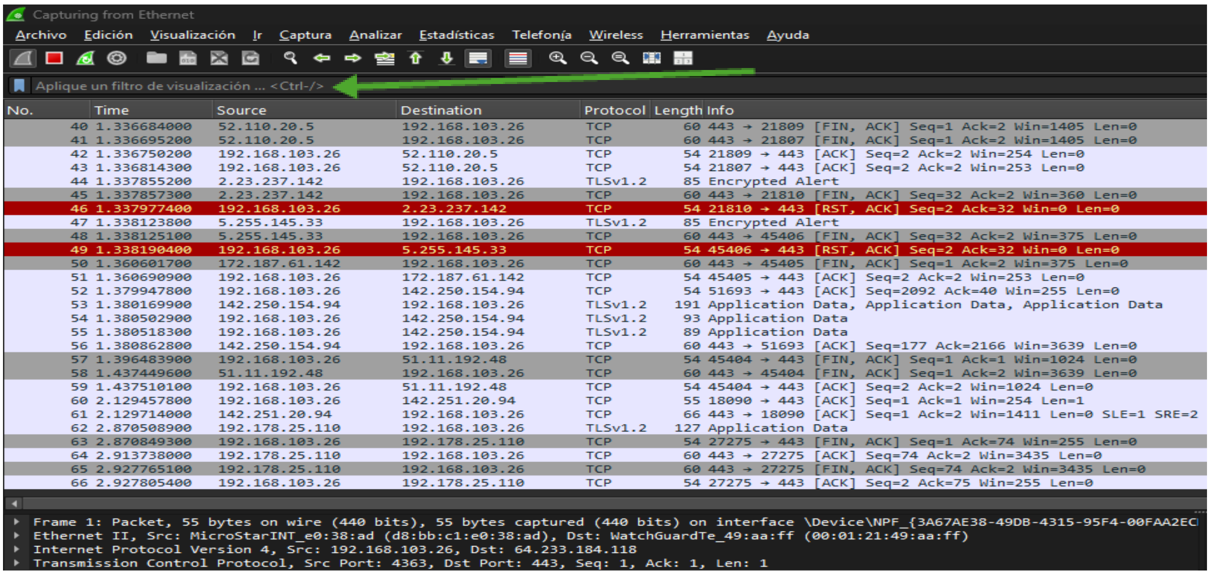
WIRESHARK

Wireshark es una herramienta de análisis de red que permite capturar y filtrar paquetes en tiempo real para estudiar el tráfico que circula por una red. Gracias a sus filtros, es posible identificar protocolos, direcciones IP, puertos, MAC y distintos tipos de comunicación, facilitando tareas de monitorización, diagnóstico y análisis de redes.

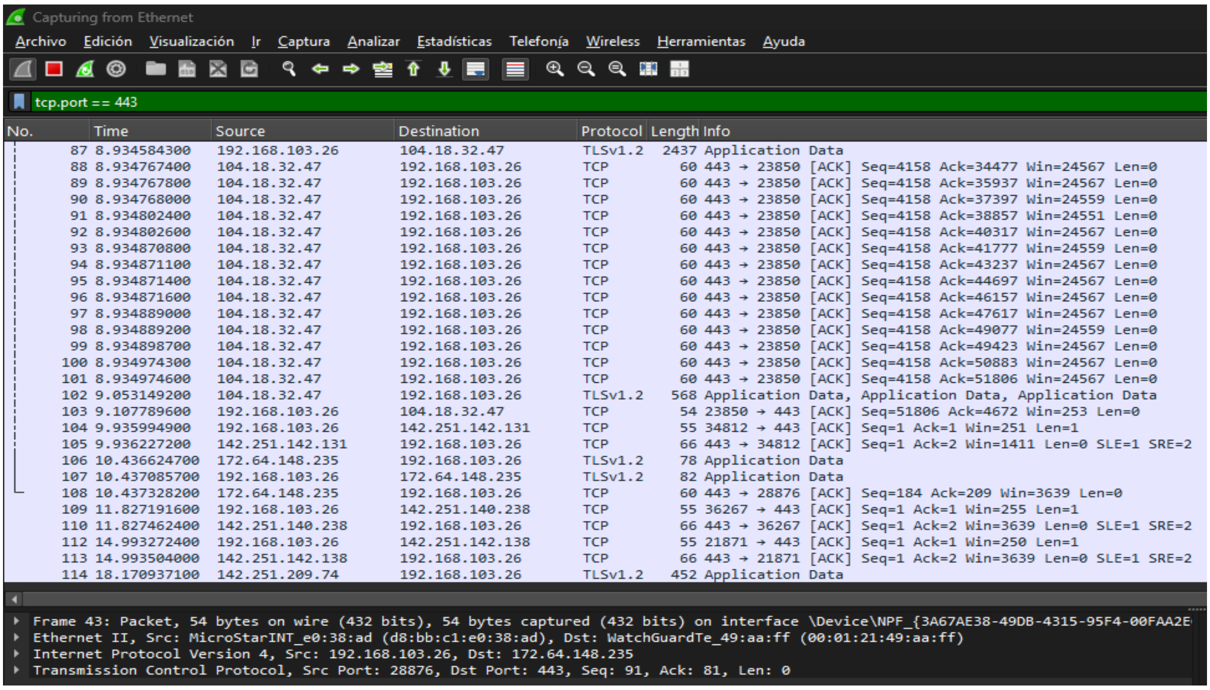
Primero en Wireshark elegimos lo que queremos capturar, en este caso Ethernet que es donde circula la mayoría de trafico



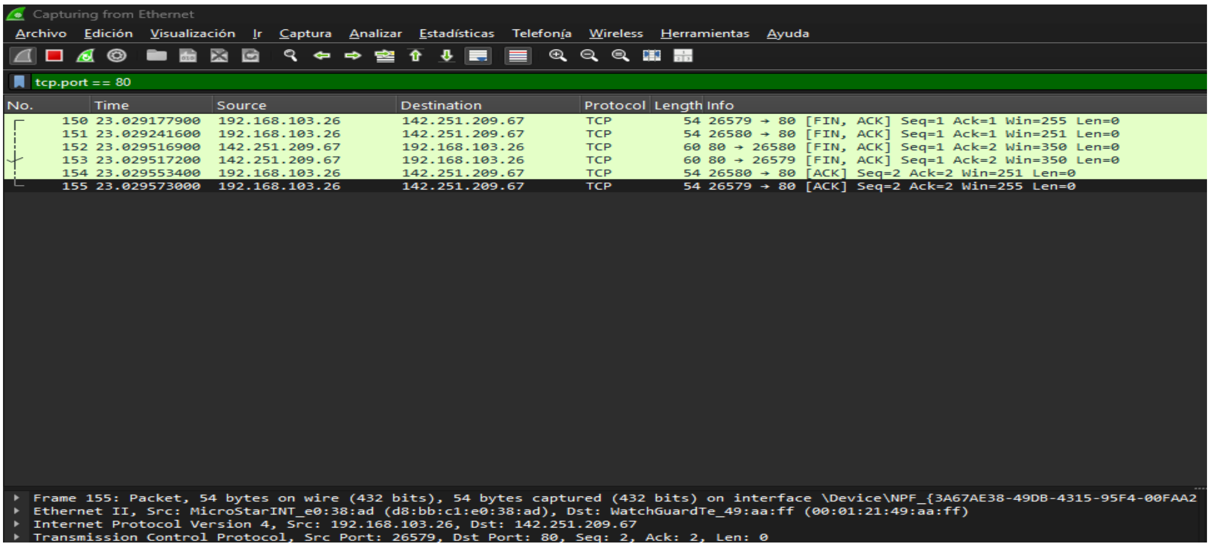
Aquí en la barra de filtros es donde vamos a escribir para filtrar lo que queremos capturar



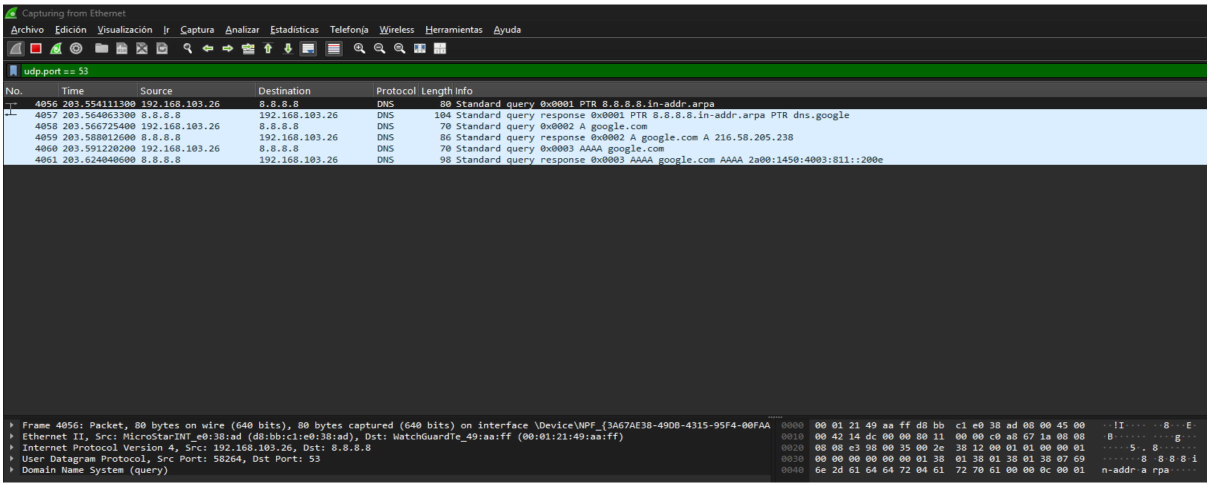
1. Deseamos visualizar todos los tcp que circulen por el puerto 443. Filtro tcp.port == 443



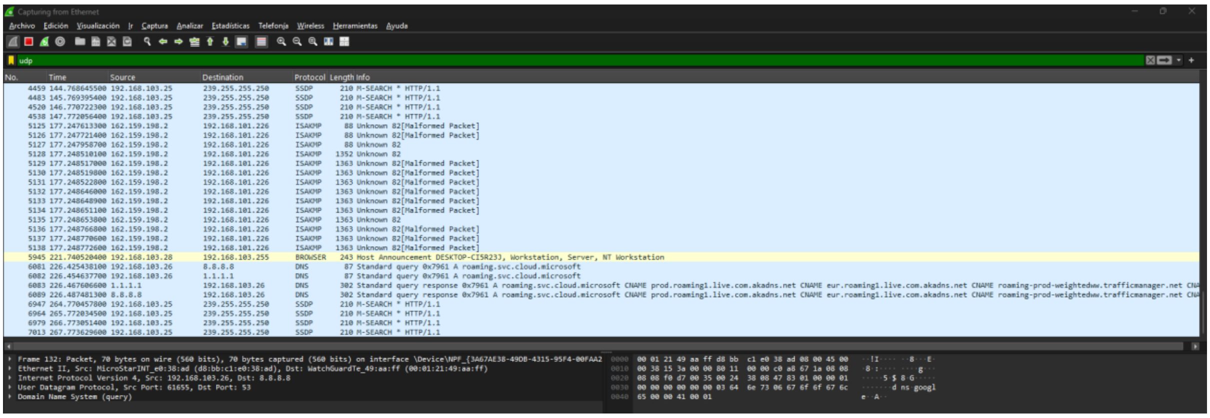
2. Localizar todos los TCP que circulen por el puerto 80. Filtro: tcp.port == 80



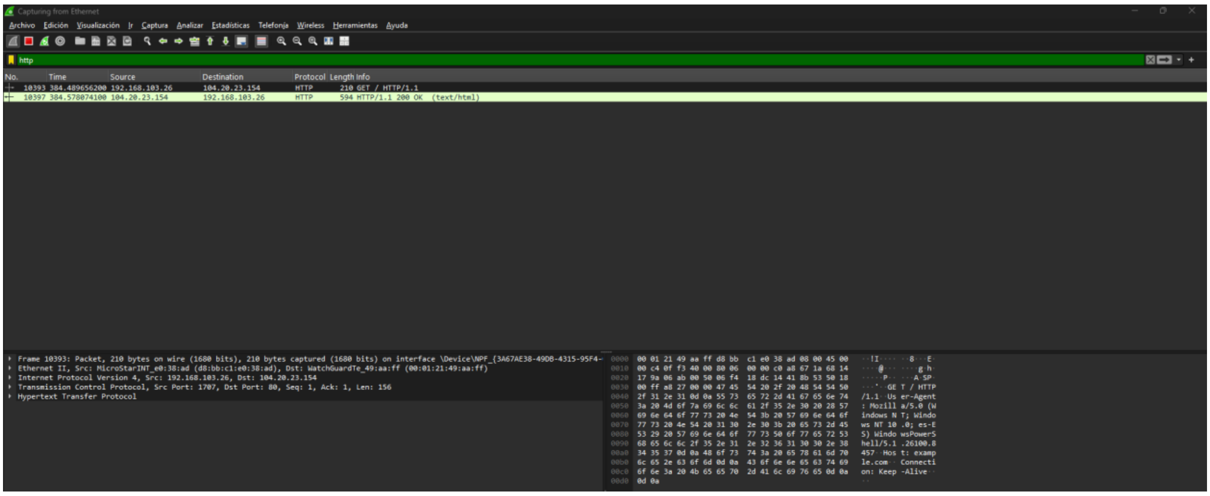
3. Localizar todos los UDP que circulen por el puerto 53. Filtro: udp.port == 53



4. Localizar todos los paquetes que circulen por UDP. Filtro: udp

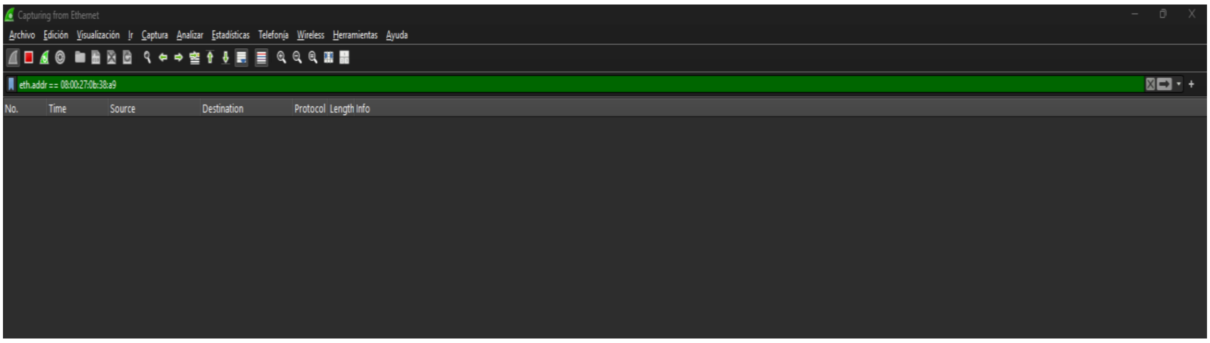


5. Orden para localizar todos los http. Filtro: http



6. Localizar la MAC 08:00:27:0b:38:a9. Filtro: eth.addr == 08:00:27:0b:38:a9

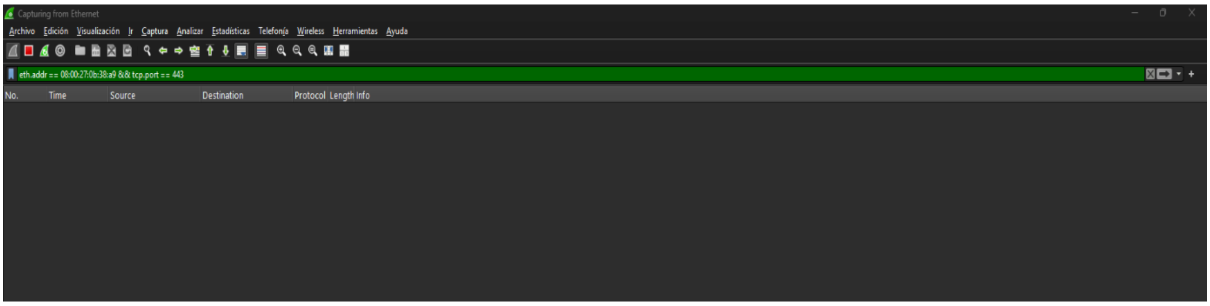
No ha dado resultado debido a que no ha encontrado tráfico asociado a esa dirección MAC en la red



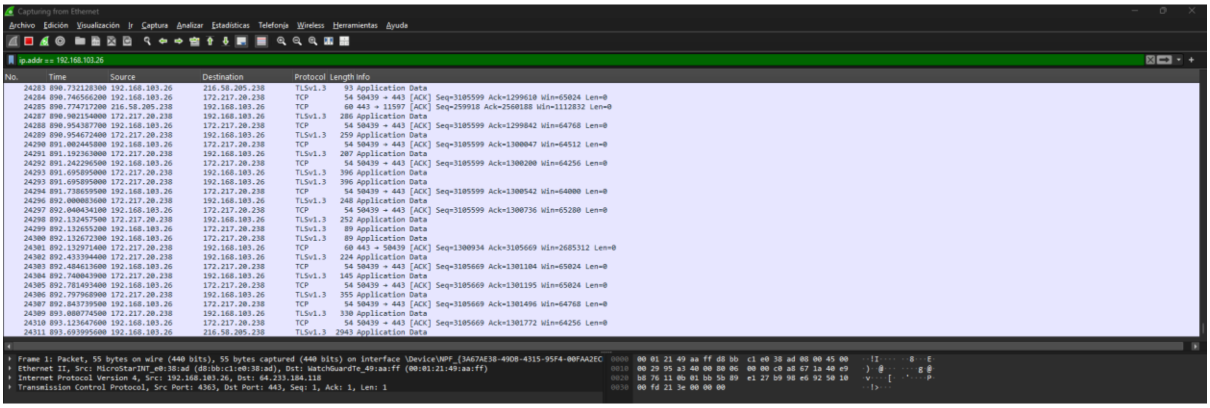
7. Localizar la MAC anterior a través de los tcp “Y” que circulen por el puerto 443.

Filtro: eth.addr == 08:00:27:0b:38:a9 && tcp.port == 443

No ha dado resultado debido a que no ha encontrado tráfico asociado a esa dirección MAC en la red

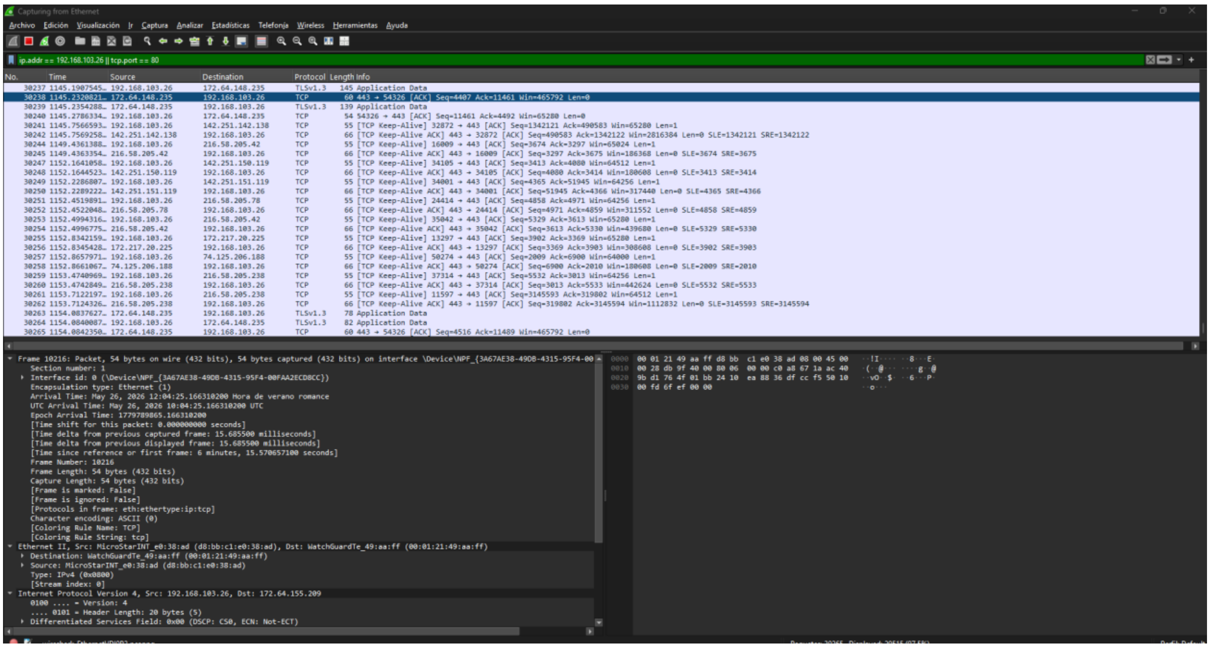


8. Localizar la ip 192.168.103.26. Filtro: ip.addr == 192.168.103.26

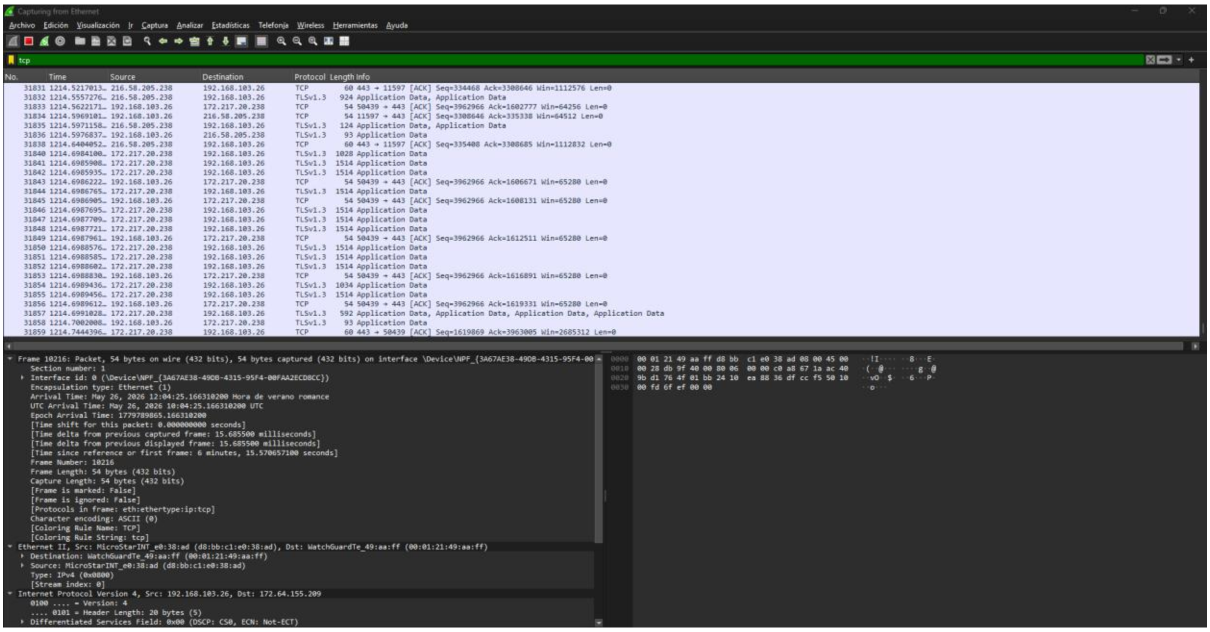


9. Localizar la ip anterior “O” que circule por el puerto 80.

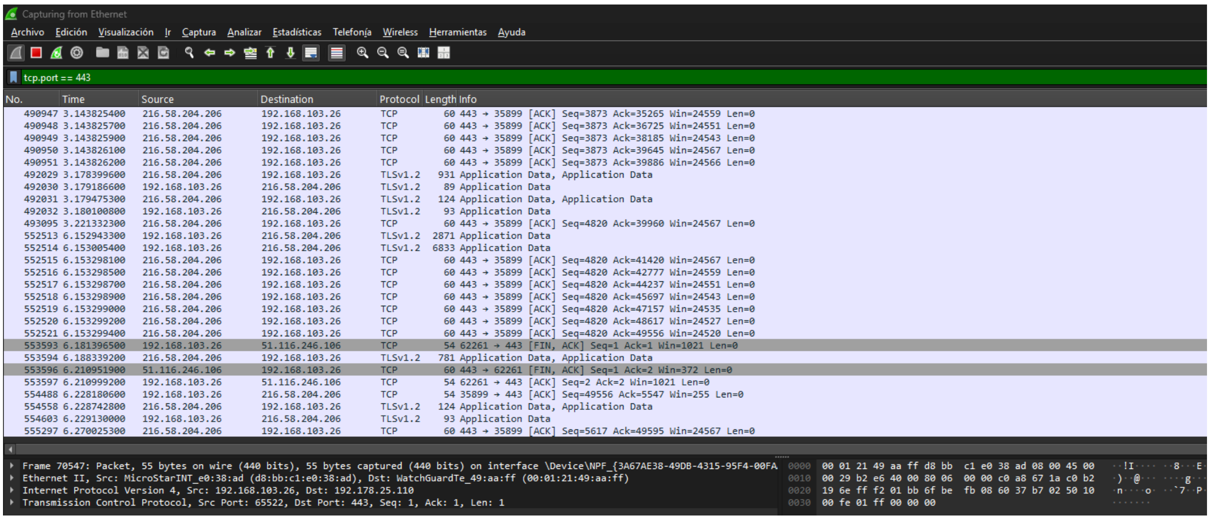
Filtro: ip.addr == 192.168.103.26 || tcp.port == 80



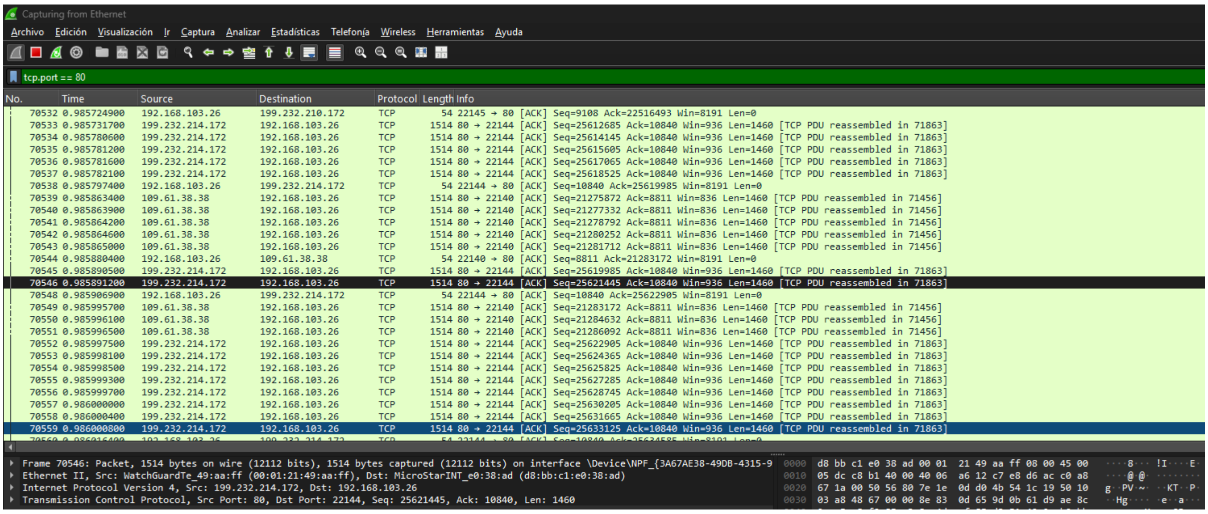
10. Localizar todo el tráfico tcp. Filtro: tcp



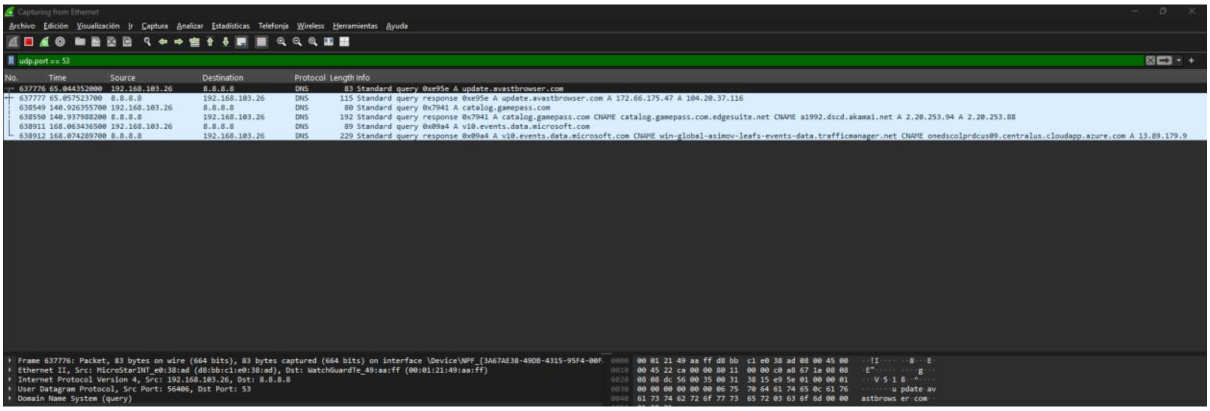
11. Localizar todo el tráfico HTTPS. Filtro: tcp.port == 443



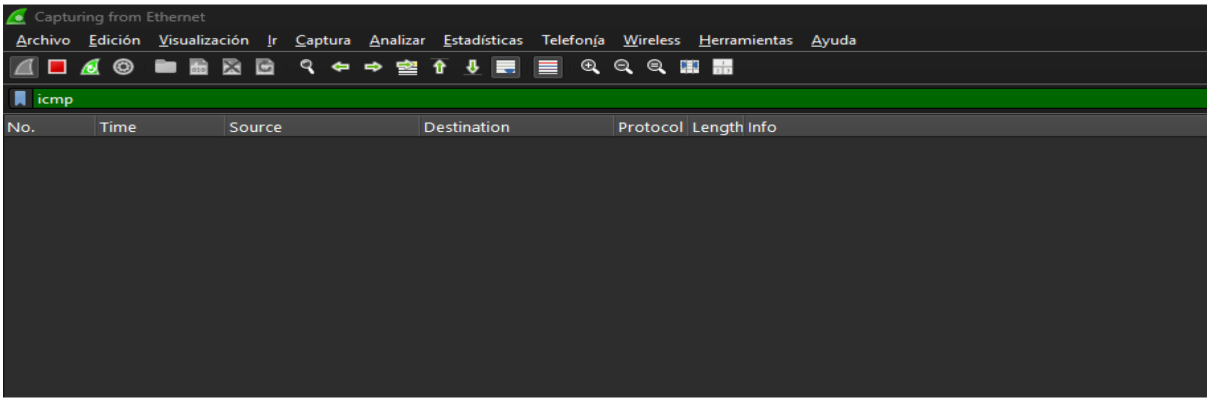
12. Localizar todo el tráfico HTTP. Filtro: tcp.port == 80



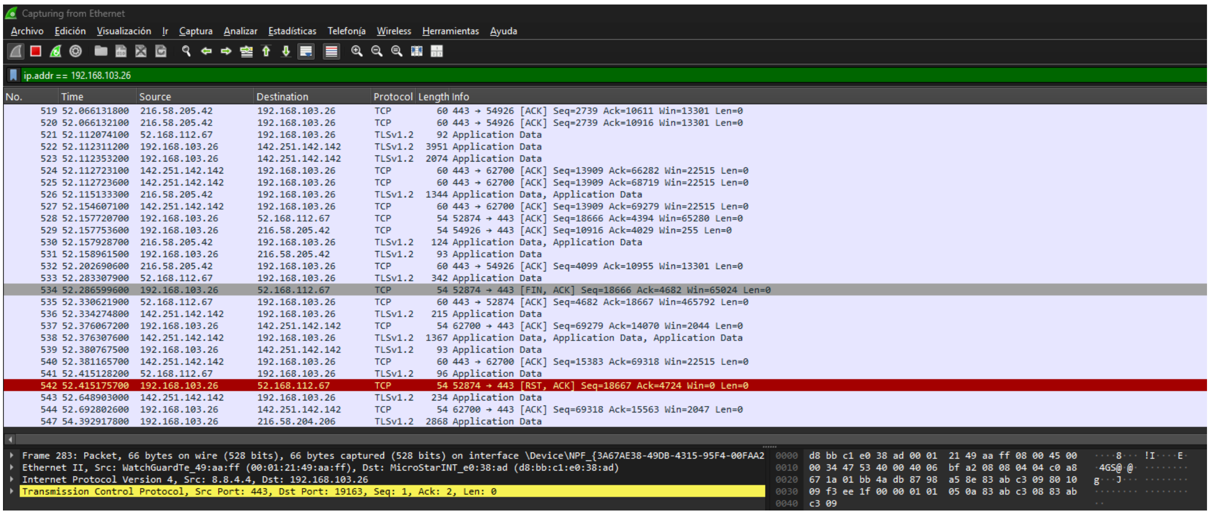
13. Mostrar únicamente tráfico DNS por UDP. Filtro: udp.port == 53



14. Buscar todos los paquetes ICMP (ping). Filtro: icmp

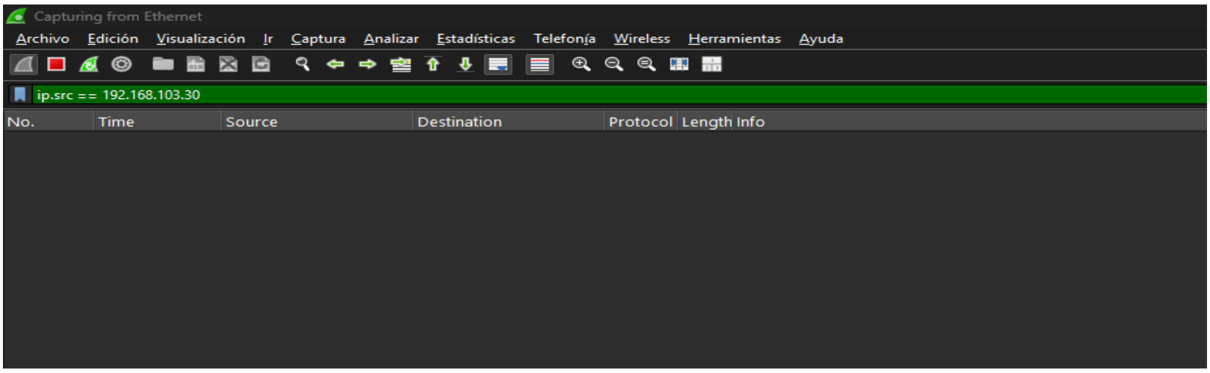


15. Localizar tráfico de la IP 192.168.103.26. Filtro: ip.addr == 192.168.103.26



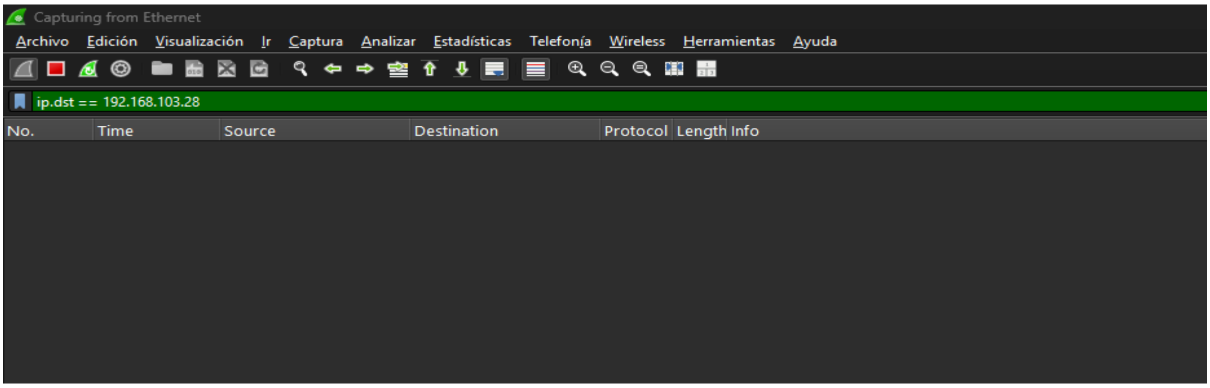
16. Mostrar únicamente tráfico origen desde la IP 192.168.103.30

Filtro: ip.src == 192.168.103.30

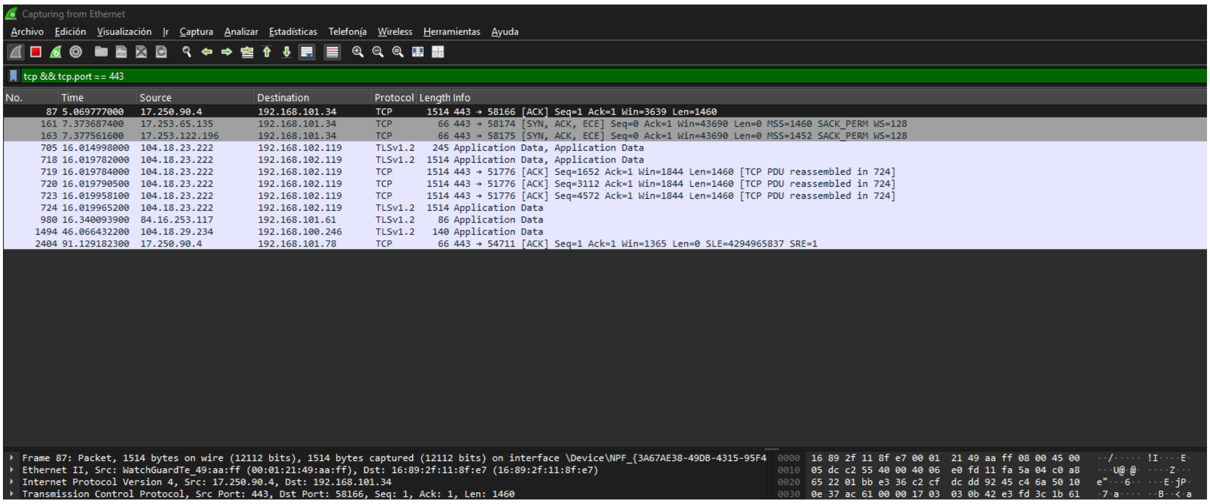


17. Mostrar únicamente tráfico destino hacia la IP 192.168.103.28

Filtro: ip.dst == 192.168.103.28

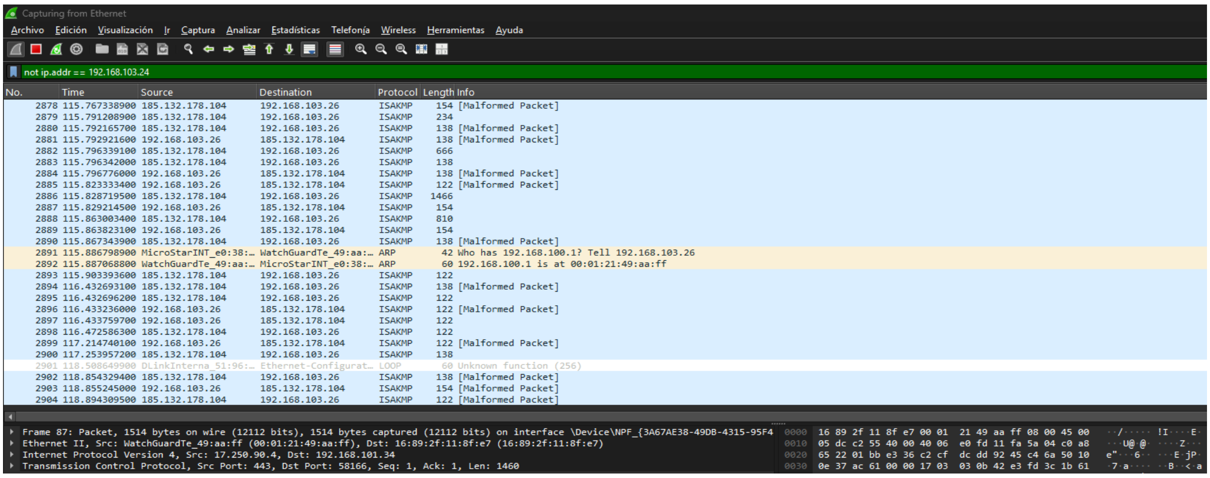


18. Buscar tráfico TCP y HTTPS al mismo tiempo. Filtro: tcp && tcp.port == 443

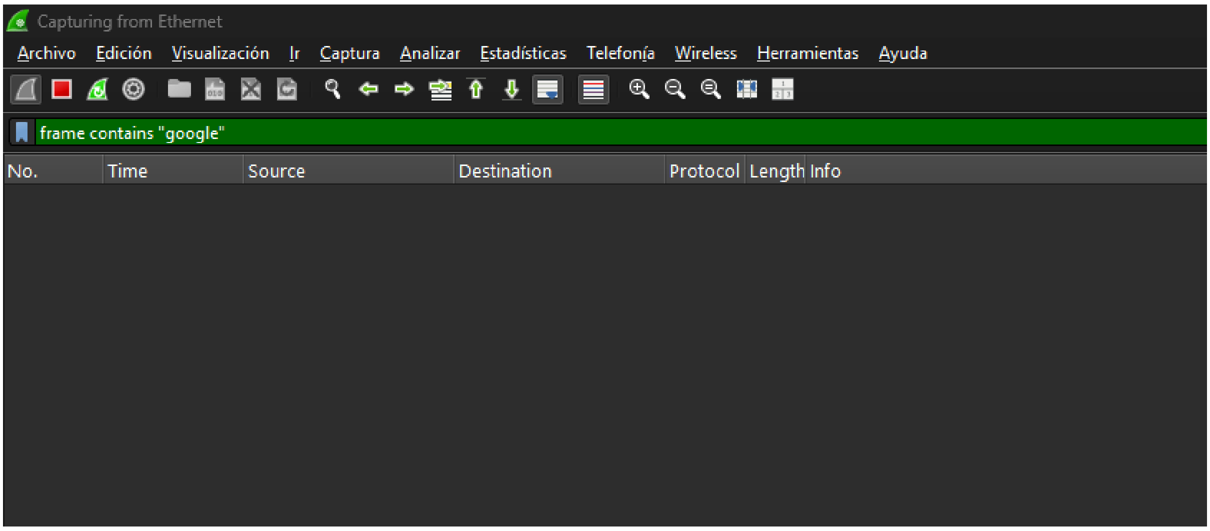


19. Localizar todos los paquetes que no pertenezcan a la IP 192.168.103.24.

Filtro: not ip.addr == 192.168.103.24



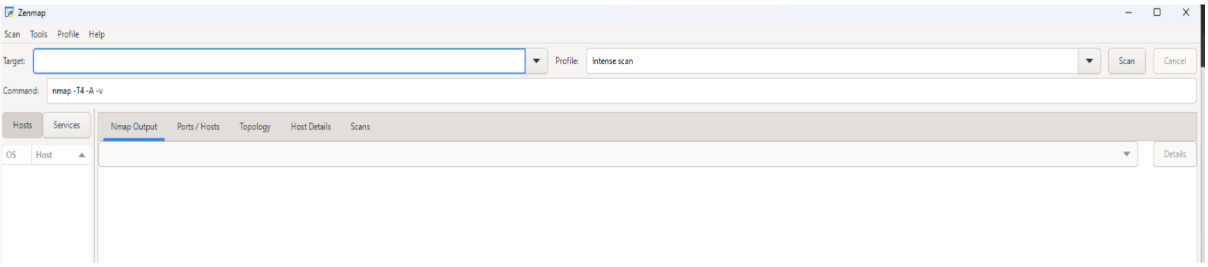
20. Buscar paquetes que contengan la palabra “google”. Filtro: frame contains "google"



NMAP

Nmap es una herramienta de escaneo y análisis de redes utilizada para detectar dispositivos conectados, puertos abiertos y servicios activos dentro de una red. Permite realizar distintos tipos de análisis, desde escaneos rápidos hasta detección de sistemas operativos y versiones de servicios, siendo una herramienta fundamental para administración y seguridad informática.

Interfaz gráfica de NMAP



Se ejecutarán los comandos desde CMD, utilizando Nmap escribiendo directamente “nmap” seguido de las opciones correspondientes en cada instrucción para realizar los distintos escaneos de red.

1. Deseamos escanear de forma silenciosa nuestra red.

Filtro: nmap -sS 192.168.103.22-33

```
C:\Program Files (x86)\Nmap>nmap -sS 192.168.103.22-33
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:20 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00085s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
111/tcp  open  rpcbind
3128/tcp open  squid-http
8000/tcp open  http-alt
9000/tcp open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0027s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
2179/tcp open  vmrdp
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.25
Host is up (0.00094s latency).
Not shown: 983 closed tcp ports (reset)
PORT     STATE SERVICE
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
3000/tcp open  ppp
8000/tcp open  http-alt
8081/tcp open  blackice-icecap
8082/tcp open  blackice-alerts
8083/tcp open  us-srv
8084/tcp open  websnp
8085/tcp open  unknown
8086/tcp open  d-s-n
8087/tcp open  simplifymedia
8088/tcp open  radan-http
8089/tcp open  unknown
8099/tcp open  unknown
9090/tcp open  zeus-admin
9100/tcp open  jetdirect
```

2. Queremos conocer la dirección IP de Adams.es. Filtro: nmap adams.es

```
C:\Program Files (x86)\Nmap>nmap adams.es
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:23 +0200
Nmap scan report for adams.es (172.67.71.32)
Host is up (0.0018s latency).
Other addresses for adams.es (not scanned): 2606:4700:20::681a:1d0 2606:4700:20::ac43:4720 2606:4700:20::681a:d0 104.
26.1.208 104.26.0.208
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE SERVICE
53/tcp   open  domain
80/tcp   open  http
443/tcp  open  https
8080/tcp open  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 13.49 seconds
```

3. Información del sistema operativo de la máquina de destino.

Filtro: nmap -O 192.168.103.25

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -O 192.168.103.25
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:24 +0200
Nmap scan report for 192.168.103.25
Host is up (0.00057s latency).
Not shown: 983 closed tcp ports (reset)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
3000/tcp open  ppp
8000/tcp open  http-alt
8081/tcp open  blackice-icecap
8082/tcp open  blackice-alerts
8083/tcp open  us-srv
8084/tcp open  websnp
8085/tcp open  unknown
8086/tcp open  d-s-n
8087/tcp open  simplifymedia
8088/tcp open  radan-http
8089/tcp open  unknown
8099/tcp open  unknown
9090/tcp open  zeus-admin
9100/tcp open  jetdirect
MAC Address: 2C:CF:67:7F:E1:B6 (Raspberry Pi (Trading))
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.99%E=4%D=5/26%OT=53%CT=1%CU=37477%PV=Y%DS=1%DC=D%G=Y%M=2CCF67%T
OS:M=6A159102%P=i686-pc-windows-windows)SEQ(SP=104%GCD=1%ISR=107%TI=Z%CI=Z%
OS:II=I%TS=21)SEQ(SP=104%GCD=1%ISR=10A%TI=Z%CI=Z%II=I%TS=21)SEQ(SP=104%GCD=
OS:1%ISR=10D%TI=Z%CI=Z%II=I%TS=22)SEQ(SP=105%GCD=1%ISR=10A%TI=Z%CI=Z%II=I%T
OS:S=21)SEQ(SP=FF%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=21)OPS(O1=M5B4ST11NW9%O2=
OS:M5B4ST11NW9%O3=M5B4NNT11NW9%O4=M5B4ST11NW9%O5=M5B4ST11NW9%O6=M5B4ST11)WI
OS:N(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=3F%W=FA
OS:F0%O=M5B4NNSNW9%CC=Y%Q=)T1(R=Y%DF=Y%T=3F%S=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3
OS:(R=N)T4(R=Y%DF=Y%T=3F%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=
OS:Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=
OS:Y%DF=Y%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%R
OS:IPL=G%RID=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.38 seconds
```

4. Escaneo rápido de IP de destino. Filtro: nmap -F 192.168.103.27

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -F 192.168.103.27
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-26 14:25 +0200
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.61 seconds
```

5. Escaneo rápido de IP de destino parecido a nmap T4. Filtro: nmap -T4 192.168.103.30

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -T4 192.168.103.30
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:20 +0200
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.62 seconds
```

6. Necesitamos escanear a la vez varias IP

Filtro: nmap 192.168.103.22 192.168.103.25 192.168.103.31

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.22 192.168.103.25 192.168.103.31
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:21 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00055s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
111/tcp  open  rpcbind
3128/tcp open  squid-http
8000/tcp open  http-alt
9000/tcp open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap done: 3 IP addresses (1 host up) scanned in 2.17 seconds
```

7. Información detallada de una IP en concreto según la va encontrando nmap.

Filtro: nmap -sV 192.168.103.24

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -sV 192.168.103.24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:22 +0200
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 1.85 seconds
```

8. Información de todos los comandos de nmap. Filtro: nmap --help

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap --help
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
```

9. Escanear solo máquinas conectadas (sin puertos). Filtro: nmap -sn 192.168.103.22-33

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -sn 192.168.103.22-33
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:25 +0200
Stats: 0:00:00 elapsed; 0 hosts completed (0 up), 11 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 68.18% done; ETC: 09:25 (0:00:00 remaining)
Nmap scan report for 192.168.103.22
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)
Nmap scan report for 192.168.103.23
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)
Nmap scan report for 192.168.103.27
Host is up (0.0010s latency).
MAC Address: E8:D8:D1:D2:F5:3A (HP)
Nmap scan report for 192.168.103.28
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:70 (Micro-Star Intl)
Nmap scan report for 192.168.103.29
Host is up (0.0010s latency).
MAC Address: 04:7C:16:00:B7:3B (Micro-Star Intl)
Nmap scan report for 192.168.103.33
Host is up (0.00s latency).
MAC Address: B8:97:5A:6F:CD:F3 (Biostar Microtech Int'l)
Nmap scan report for 192.168.103.26
Host is up.
Nmap done: 12 IP addresses (7 hosts up) scanned in 3.43 seconds
```

10. Escanear los primeros 500 puertos de una IP en concreto.

Filtro: nmap -p 1-500 192.168.103.28

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -p 1-500 192.168.103.28
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:26 +0200
Nmap scan report for 192.168.103.28
Host is up (0.0015s latency).
Not shown: 497 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
MAC Address: 04:7C:16:00:B7:70 (Micro-Star Intl)

Nmap done: 1 IP address (1 host up) scanned in 1.84 seconds
```

11. Informarse de la IP de Google.es. Filtro: nmap google.es

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap google.es
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:27 +0200
Nmap scan report for google.es (172.217.171.35)
Host is up (0.0023s latency).
Other addresses for google.es (not scanned): 2a00:1450:4003:811::2003
rDNS record for 172.217.171.35: lcmadb-ac-in-f3.1e100.net
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE
53/tcp  open  domain
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 9.05 seconds
```

12. Conocer los primeros 800 puertos de Google.es. Filtro: nmap -p 1-800 google.es

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -p 1-800 google.es
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:27 +0200
Nmap scan report for google.es (172.217.171.35)
Host is up (0.0021s latency).
Other addresses for google.es (not scanned): 2a00:1450:4003:804::2003
rDNS record for 172.217.171.35: lcmadb-ac-in-f3.1e100.net
Not shown: 797 filtered tcp ports (no-response)
PORT    STATE SERVICE
53/tcp  open  domain
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 12.97 seconds
```

13. Escanear toda la red. Filtro: nmap 192.168.103.0/24

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:28 +0200
Nmap scan report for 192.168.103.22
Host is up (0.0015s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
111/tcp  open  rpcbind
3128/tcp open  squid-http
8000/tcp open  http-alt
9000/tcp open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0077s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
2179/tcp open  vmrdp
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.0029s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
912/tcp  open  apex-mesh
```

14. Analizar varias máquinas (último octeto). Filtro: nmap 192.168.103.24,26,33

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.24,26,33
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:29 +0200
Nmap scan report for 192.168.103.33
Host is up (0.0028s latency).
Not shown: 982 filtered tcp ports (no-response)
PORT     STATE SERVICE
21/tcp   open  ftp
53/tcp   open  domain
80/tcp   open  http
88/tcp   open  kerberos-sec
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
143/tcp  open  imap
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
587/tcp  open  submission
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
3389/tcp open  ms-wbt-server
5357/tcp open  wsdapi
5985/tcp open  wsman
MAC Address: B8:97:5A:6F:CD:F3 (Biostar Microtech Int'l)

Nmap scan report for 192.168.103.26
Host is up (0.000088s latency).
```

15. Escanear el rango de direcciones. Filtro: nmap 192.168.103.20-50

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.20-50
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:30 +0200
Nmap scan report for 192.168.103.22
Host is up (0.0015s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
111/tcp  open  rpcbind
3128/tcp open  squid-http
8000/tcp open  http-alt
9000/tcp open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.015s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
912/tcp  open  apex-mesh
2179/tcp open  vmrdp
3389/tcp open  ms-wbt-server
5357/tcp open  wsdapi
MAC Address: E8:D8:D1:D2:F5:3A (HP)

Nmap scan report for 192.168.103.28
Host is up (0.0018s latency).
Not shown: 994 closed tcp ports (reset)
```

16. Escanear la red excluyendo una IP.

Filtro: nmap 192.168.103.0/24 --exclude 192.168.103.25

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.0/24 --exclude 192.168.103.25
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:31 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00073s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
111/tcp  open  rpcbind
3128/tcp open  squid-http
8000/tcp open  http-alt
9000/tcp open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0011s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
2179/tcp open  vmrdp
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.0011s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
```

17. Escanear la red excluyendo un rango de IPs.

Filtro: nmap 192.168.103.0/24 --exclude 192.168.103.24-28

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.0/24 --exclude 192.168.103.24-28
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:31 +0200
Nmap scan report for 192.168.103.22
Host is up (0.0018s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
111/tcp  open  rpcbind
3128/tcp open  squid-http
8000/tcp open  http-alt
9000/tcp open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.29
Host is up (0.00089s latency).
Not shown: 993 closed tcp ports (reset)
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
912/tcp  open  apex-mesh
3306/tcp open  mysql
5357/tcp open  wsdapi
MAC Address: 04:7C:16:00:B7:3B (Micro-Star Intl)

Nmap scan report for 192.168.103.30
Host is up (0.00076s latency).
```

18. Qué hace la orden nmap 192.168.103.*. Filtro: escanea toda la red 192.168.103.0/24

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap 192.168.103.*
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:32 +0200
Nmap scan report for 192.168.103.22
Host is up (0.00087s latency).
Not shown: 995 closed tcp ports (reset)
PORT     STATE SERVICE
22/tcp   open  ssh
111/tcp  open  rpcbind
3128/tcp open  squid-http
8000/tcp open  http-alt
9000/tcp open  cslistener
MAC Address: 04:7C:16:00:B7:7E (Micro-Star Intl)

Nmap scan report for 192.168.103.23
Host is up (0.0023s latency).
Not shown: 998 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
2179/tcp open  vmrdp
MAC Address: 04:7C:16:00:B7:3C (Micro-Star Intl)

Nmap scan report for 192.168.103.27
Host is up (0.0015s latency).
Not shown: 993 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
445/tcp  open  microsoft-ds
902/tcp  open  iss-realsecure
```

19. Conocer la versión de nmap. Filtro: nmap -V

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap -v
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:33 +0200
Read data files from: C:\Program Files (x86)\Nmap
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.05 seconds
           Raw packets sent: 0 (0B) | Rcvd: 0 (0B)
```

20. Servidores de prueba Nmap. Filtro: scanme.nmap.org y testphp.nmap.org

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap scanme.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:34 +0200
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.16s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 996 filtered tcp ports (no-response)
PORT     STATE  SERVICE
53/tcp   open   domain
80/tcp   open   http
443/tcp  open   https
8080/tcp closed http-proxy

Nmap done: 1 IP address (1 host up) scanned in 28.45 seconds
```

```
Administrador: Símbolo del sistema

C:\Program Files (x86)\Nmap>nmap testphp.nmap.org
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-27 09:35 +0200
Nmap scan report for testphp.nmap.org (50.116.1.184)
Host is up (0.0013s latency).
Other addresses for testphp.nmap.org (not scanned): 2600:3c01:e000:3e6::6d4e:7061
rDNS record for 50.116.1.184: ack.nmap.org
Not shown: 997 filtered tcp ports (no-response)
PORT    STATE SERVICE
53/tcp  open  domain
80/tcp  open  http
443/tcp open  https

Nmap done: 1 IP address (1 host up) scanned in 14.92 seconds
```

BIBLIOGRAFIA

Gordon Lyon, F. (2025). *Nmap Network Scanning*. Nmap Project.

https://nmap.org/book/

Nmap Project. (2025). *Nmap Reference Guide*.

https://nmap.org/docs.html

Wireshark Foundation. (2025). *Wireshark User Guide*.

https://www.wireshark.org/docs/wsug_html_chunked/

Wireshark Foundation. (2025). *Wireshark Documentation*.

https://www.wireshark.org/docs/